

DIABETIC RETINOPATHY SCREENING REPORT

Smart India Hackathon 2026 | Problem 26038 | MathWorks AI Pipeline

Image: sample_fundus

Date: 09-Sep-2026 09:50

Quality: BORDERLINE (2/3)

1 AI DIAGNOSIS

DIAGNOSIS

Moderate NPDR

ICDR Severity Level 2 / 4 Confidence 92.8%

CONFIDENCE

92.8%

URGENT REFERRAL — Ophthalmologist consult within 7 days

2 IMAGE QUALITY METRICS

Sharpness (Laplacian Variance) Threshold: ≥ 50	19.3	FAIL
Brightness (Mean Intensity) Threshold: 30 – 210	51.3	PASS
Contrast (Std Deviation) Threshold: ≥ 25	31.3	PASS

3 DETECTED LESION EVIDENCE

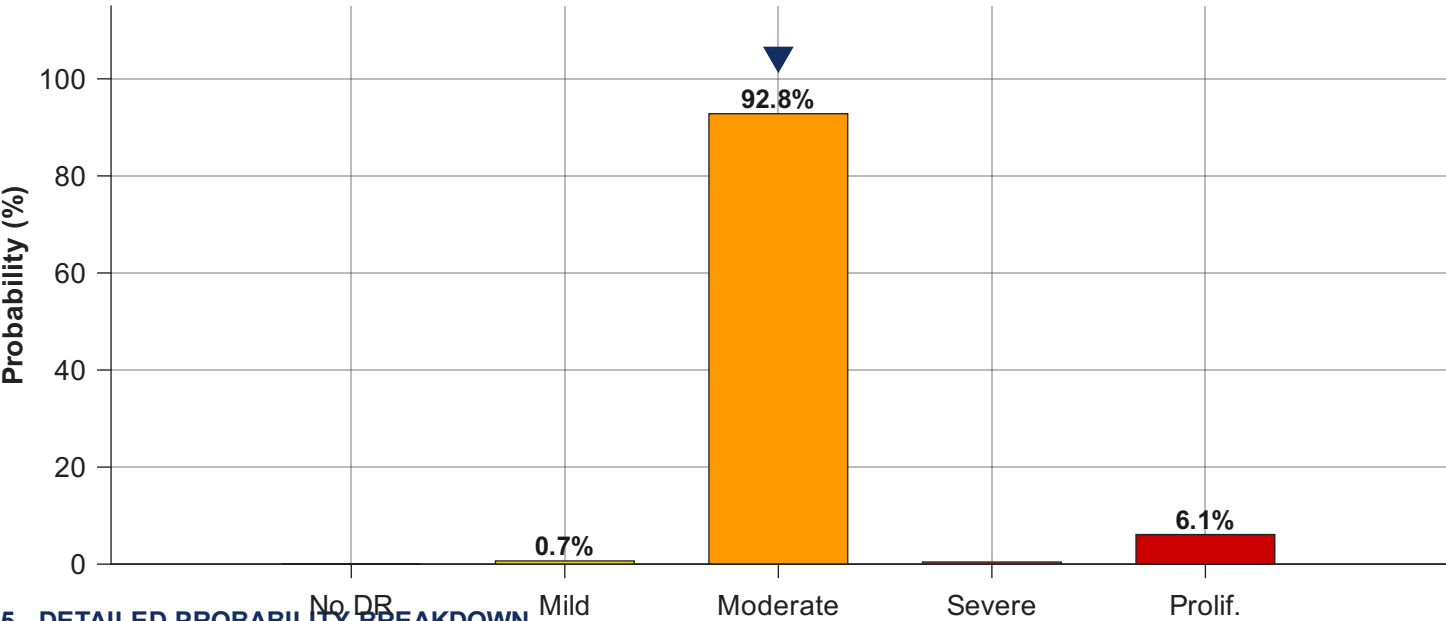
Microaneurysms (MA) Earliest DR sign — small vascular microleaks	24 detected
Hard Exudates (EX) Lipid deposits — blood-retinal barrier breakdown	1 detected
Hemorrhages (HE) Deeper vascular wall ruptures	0 detected

Optic Disc [371, 239] Fovea [454, 244]

CLASS PROBABILITY ANALYSIS

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4 PROBABILITY DISTRIBUTION



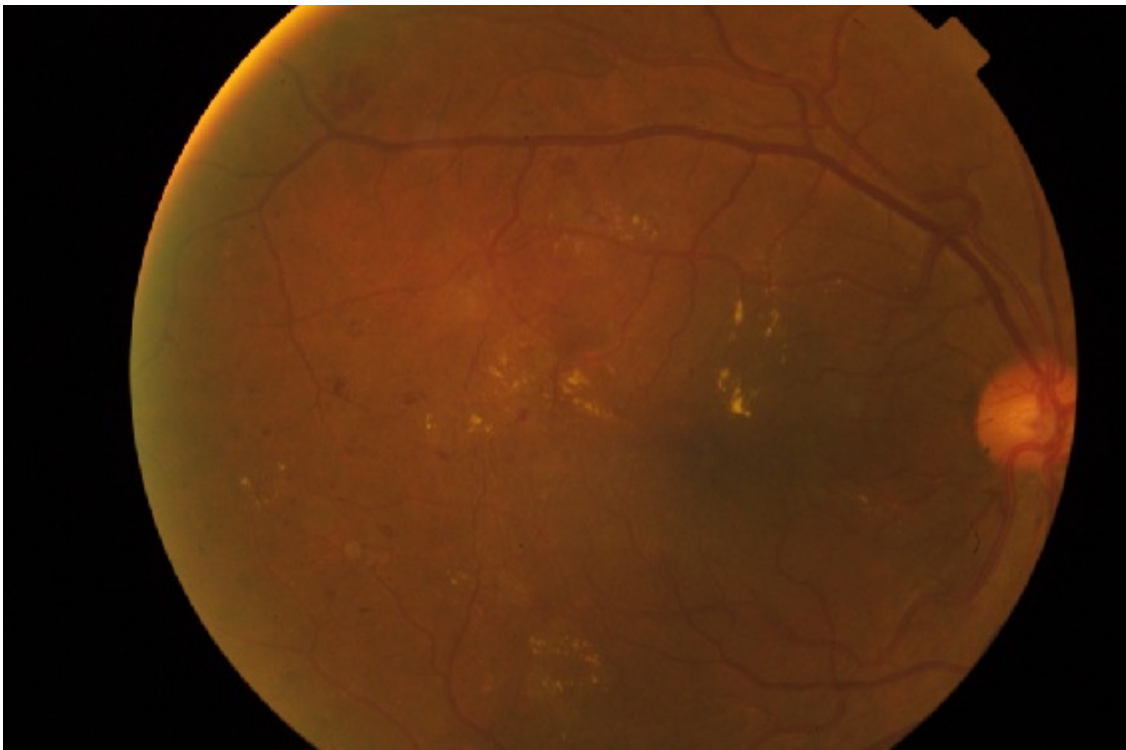
5 DETAILED PROBABILITY BREAKDOWN

No DR Level 0 / 4		0.0%
Mild NPDR Level 1 / 4		0.7%
Moderate NPDR Level 2 / 4		92.8% AI
Severe NPDR Level 3 / 4		0.4%
Proliferative DR Level 4 / 4		6.1%

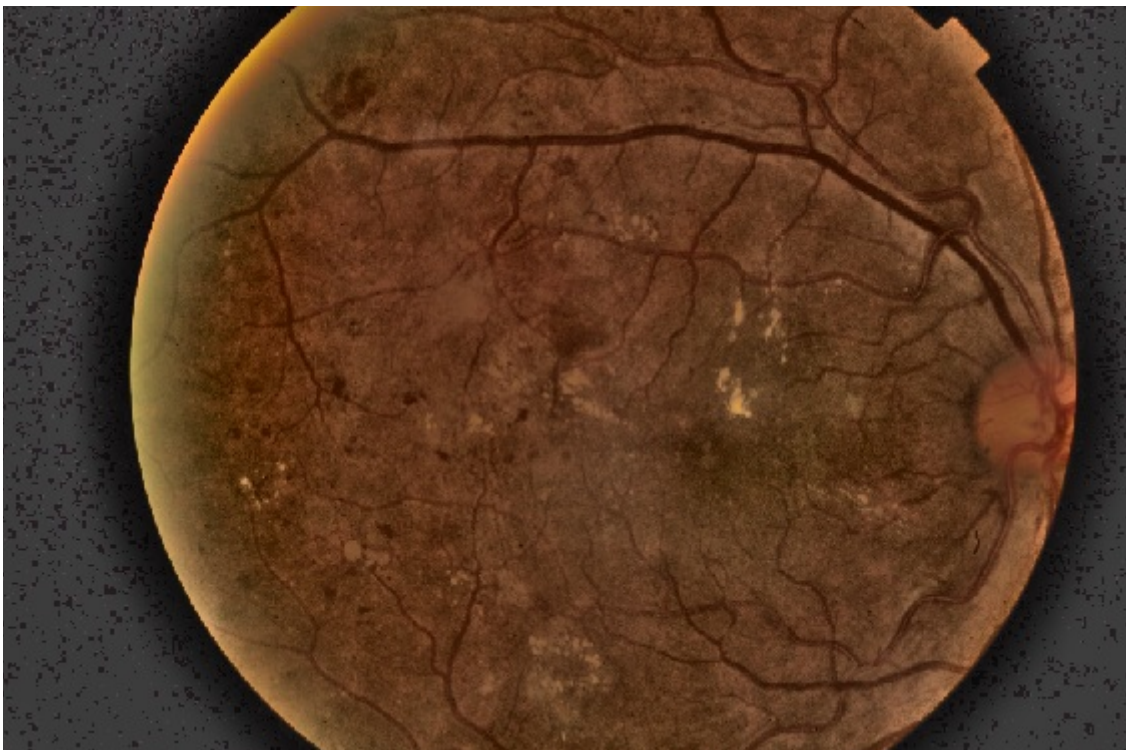
FUNDUS IMAGE QUALITY & ENHANCEMENT

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A ORIGINAL FUNDUS IMAGE



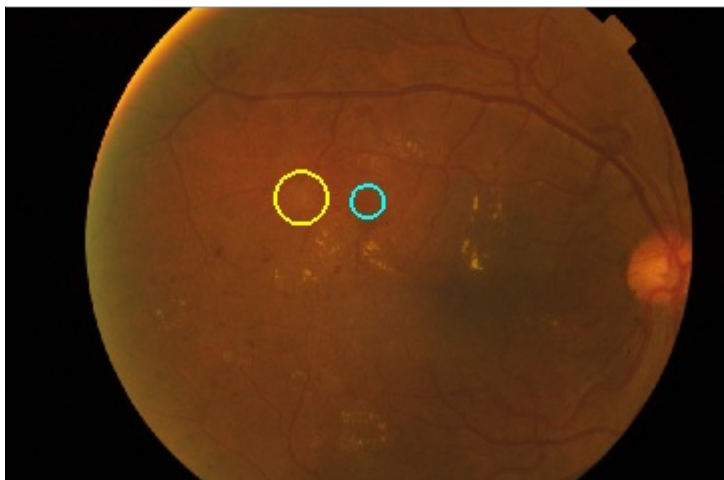
B CLAHE ENHANCED & ILLUMINATION NORMALISED



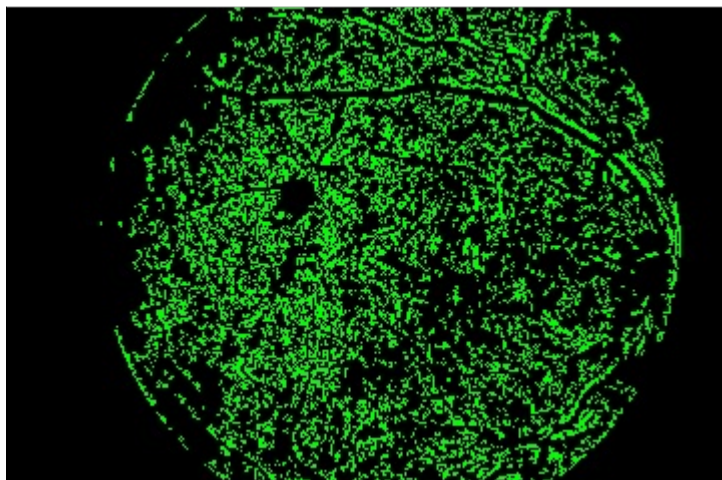
RETINAL SEGMENTATION & EXPLAINABLE AI

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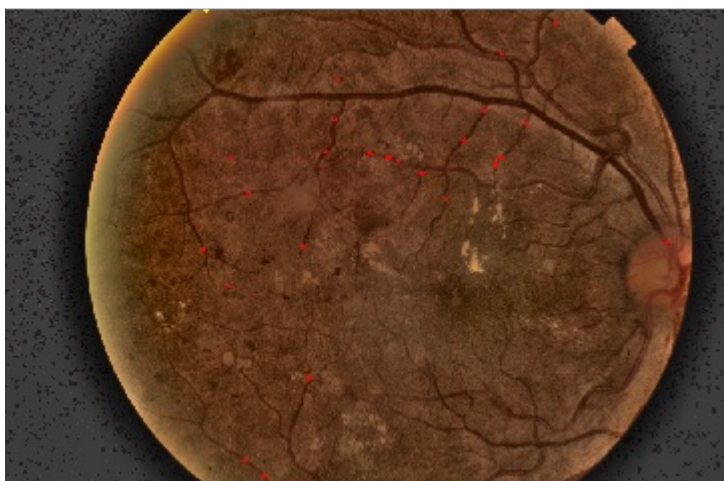
C. Anatomical Landmarks (OD=Yellow, Fovea=Cyan)



D. Vessel Network Segmentation



E. Lesion Map (MA=24, EX=1, HE=0)



F. Grad-CAM Explainability (92.8% Confidence)

